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School-Based Social Skills Training for Children With Autism Spectrum Disorder

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Abstract

The present study evaluated the effects of the Superheroes Social Skills program, a social skills curriculum for children with autism spectrum disorders (ASD). Previous research has found the curriculum to improve social engagements of children with ASD during unstructured recess periods but has been limited in research design and lack of maintenance data. Five elementary-age participants with current placements in inclusive public school settings were included in the study. A multiple baseline design across participants was used to determine the effects of the intervention on social engagement during unstructured recess periods, as well as teacher ratings of social functioning and sociometric status. Visual and statistical analyses of data indicate increased social engagement of participants during recess periods following introduction of social skills training. Improvements were also observed in teacher ratings of social functioning and sociometric status. Limitations of the current study are also discussed.

Keywords

social skills, socialization, elementary, peer modeling, social, skills

Although individuals with autism spectrum disorders (ASD) vary in their individual symptoms and presentation, ASD is marked by impairments in social interactions and communication and restrictive and repetitive behavior (American Psychiatric Association, 2013). Common social problems include initiating and maintaining eye contact, conversations, and reciprocal behavior. Given that these deficits in social behavior can have long-lasting effects on the development of appropriate relationships with others (e.g., Bauminger, Shulman, & Agam, 2003), it is no surprise that social skills are commonly targeted on individualized education plans (IEPs) and other school-based intervention plans (Weiss & Harris, 2001). Findings indicate that social skills are addressed in 80% of IEPs for children with ASD (Ruble, McGrew, Dalrymple, & Jung, 2010). Despite the frequency with which social skills deficits are addressed within school settings, evidence-based social skills training is rarely delivered in this context, with social skills training often consisting of unsupported intervention approaches (e.g., Bellini, Peters, Benner, & Hopf, 2007; Hess, Morrier, Heflin, & Ivey, 2008). In addition, Bellini, Peters, et al. (2007) indicated that the common practice of conducting school-based social skills training in pullout groups decreases school-based social skills training efficacy. Given the frequent use of nonempirically supported interventions and intervention that takes in pullout groups, it is unsurprising that meta-analysis of school-based social skills training has found school-based intervention to result in only moderate maintenance and minimal generalization effects (Bellini, Peters, et al., 2007).

The typical method for promoting social skills is to teach individuals discrete social skill steps via instructor and student role-play (Elliott, Roach, & Beddow, 2008). Although this method of instruction may produce improvements in demonstration of the target skill within the training environment, didactic instruction of discrete skills to children with ASD often falls short of promoting generalized skill use, or the utilization of target social skills in novel situations (Gresham, Sugai, & Horner, 2001). To improve upon the minimal effects of social skills training within school settings (e.g., Bellini, Peters, et al., 2007) and enhance the ability of children with ASD to develop and maintain friendships, intervention strategies that both reinforce skill use and promote generalization must be implemented. Given that generalization of social skills is increasingly important as children with ASD mature and are required to

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engage in a wider variety of social settings (e.g., education, employment, independent living), it is critical that these children receive evidence-based social skills training within school settings to promote social competence, defined as the ability to use social skills in the appropriate context (Gresham & Elliott, 1987; Nuernberger, Ringdahl, Vargo, Crumpecker, & Gunnarsson, 2013). Several evidence-based strategies for promoting generalized use of target social skills have been identified, including the inclusion of typically developing peers in session, video modeling, and self-monitoring.

Inclusion of typically developing peers has been found to be a critical component of social skills training. Meta-analysis has found inclusion of typically developing peers to promote joint attention, communication, imitations, the ability to maintain interactions, and turn-taking behaviors in children with ASD (Chan et al., 2009). Most importantly, inclusion of typically developing peers has been found to result in generalized use of target social skills in nontraining environments (Zhang & Wheeler, 2011). Incorporation of typically developing peers in social skills training allows children with ASD to practice target skills with peers they encounter in authentic social settings and to access naturally occurring reinforcement from those peers. Given that access to reinforcement is a cornerstone of Stokes and Osnes's (1989) theory of generalization promotion, the inclusion of typically developing peers should be considered a key component of social skills programs.

Stokes and Osnes (1989) also identified the need for multiple training situations and behavioral exemplars to aid in promoting generalization. The use of video models, in which an individual watches a video of a target behavior and then imitates the observed model, has been found to be a useful method for promoting social skill use (e.g., Wang & Spillane, 2009), and when utilized in conjunction with didactic instruction and role-play, allows social skills training to meet Stokes and Osnes's (1989) criteria for diverse training. Video modeling has been utilized to promote acquisition of a variety of behaviors in individuals with ASD, ranging from word identification (Morlock, Reynolds, Fisher, & Comer, 2015) to purchasing behaviors (Yakubova & Taber-Doughty, 2013), as well as social engagement behaviors (e.g., Bellini, Akullian, & Hopf, 2007). Video modeling interventions are particularly promising for promoting social competence, producing generalization of target skills across settings following intervention (Bellini et al., 2007).

Self-monitoring procedures have also been found to be an important component for inclusion in social skills training (Lee, Simpson, & Shogren, 2007). A review of self-monitoring interventions targeting social skill use in children with ASD found that the intervention strategy produces increases in frequency of target skill use (Chen, 2006). Meta-analysis also indicates the utility of self-monitoring in increasing target social skills use in individuals with ASD (Lee et al., 2007). Furthermore, self-monitoring procedures allow for individuals to engage functional mediators of behavior (Radley, O'Handley, et al., 2014), the third cornerstone of Stokes and Osnes's (1989) theory for promoting generalization.

Although the incorporation of peer models, video modeling, and self-monitoring have been empirically validated for their use in promoting skill usage and generalization in individuals with ASD, they are underrepresented in school-based social skill intervention plans (Bellini & McConnell, 2010; Hess et al., 2008; Shernoff, Kratochwill, & Stoiber, 2003). Instead, school-based practitioners report implementing interventions with limited empirical support with greater frequency (Hess et al., 2008). Limited use of empirically supported interventions, such as peer modeling, video modeling, and self-monitoring may be underutilized in school settings due to limited practitioner exposure to evidence-based practice, limited time and resources, or lack of administrator support (Collier-Meek, Fallon, Johnson, Sanetti, & Delcampo, 2011). In addition, Dingfelder and Mandell (2011) suggested that evidence-based practices without an accompanying curriculum may impede adoption of such practices within school settings. As such, it is critical that practitioner-friendly social skills training curricula which incorporate evidence-based practices be developed for use by school personnel engaged in social skills training for children with ASD. Although the need for such intervention packages is apparent, there are few comprehensive social skills packages available for use within the school setting (Lord et al., 2005).

The Superheroes Social Skills program (Jenson et al., 2011) was developed to address the need for comprehensive school-based social skills interventions requiring minimal specialized training and capable of being implemented by any school personnel who typically implement intervention targeting social skills (e.g., special education teachers, school psychologists, paraprofessionals). The program incorporates several evidence-based strategies to promote skill usage and generalization, including the inclusion of typically developing peers, video models of skill usage presented via DVD, and incorporation of self-monitoring procedures for use in training and nontraining environments. The manualized intervention also utilizes animated characters to teach the skill steps to increase participant interest (e.g., Milne, Luerssen, Lewis, Leibbrandt, & Powers, 2011), facilitator and student role-play sessions, social stories in the form of comic books, and social games that make use of the target social skill. As Boardman, Arguelles, Vaughn, Hughes, and Klingner (2005) indicated that school professionals often select interventions based on feasibility, the manualized intervention may also promote greater use of evidence-based practices within school settings (e.g., Hess et al., 2008) and allow for use of the intervention by all school personnel without the need for specialized training.

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Several studies have investigated the utility of the Superheroes Social Skills program in promoting social skill use in children with ASD. In a clinic-based investigation of the Superheroes Social Skills program, Radley, Jenson, Clark, and O'Neill (2014) evaluated the effectiveness of the intervention in increasing the duration of social engagement of five children with ASD. Utilizing a multiple baseline design across two social skills groups, the researchers trained parents to facilitate all social skills training procedures. Results of the study indicated moderate improvements in duration of social engagement of participants during a free-play period ($d = 0.64$). In addition, results of the study indicated decreased reports of parental stress following termination of the social skills intervention.

In an additional evaluation of the program when used within a clinic setting, Radley, O'Handley, et al. (2014) assessed the effect of the intervention on accurate demonstration of target social skills. Utilizing a multiple baseline across skills design, the authors trained three participants with ASD in four target social skills. Following introduction of training, rapid improvements in target skill accuracy were observed for all participants. Improvements in target skill accuracy were also observed in probes of skill generalization across persons and stimuli. Overall, effects of the intervention on skill accuracy in both training and generalized settings ranged from moderate to strong (nonoverlap of all pairs [NAP] = 0.83–1.00). In addition to improvements in skill accuracy, parent-completed social skills checklists indicated improved social functioning, further suggesting generalization of skills across persons and settings.

As previous research has identified the need for curriculum-based social skills interventions to be used in school settings (Dingfelder & Mandell, 2011), several studies have evaluated the effects of the program when implemented within the context of a school. Block, Radley, Jenson, Clark, & O'Neill (2015) conducted a pilot investigation of the program using a single case, replicated AB design. Four children with ASD attended twice weekly social skills training over an 11-week period. The primary dependent variable of the study was duration of social engagement of children with ASD during recess periods, with results indicating moderate to large intervention effects (NAP = 0.78–1.00). Similar to studies conducted in clinic settings, teacher-completed social skills checklists indicated significant improvements in social functioning of participants following participation in the intervention.

Despite positive findings of Block et al. (2015), use of a replicated AB design limited the conclusions regarding efficacy of the intervention. To address the limitation of Block et al., Radley, Ford, Battaglia, and McHugh (2014) utilized a multiple baseline across participants design to assess the efficacy of the intervention in increasing social engagements of participants with ASD. Four participants with ASD participated in a once-weekly social skills group for 8 weeks. Following implementation of the intervention, each of the four participants demonstrated strong improvement in social engagement behavior during generalized recess periods (NAP = 0.93–1.00). Social skills checklists completed by parents indicated improvements in social functioning at home, further suggesting generalized use of trained skills. The researchers also utilized the Friendship Survey (Kasari, Rotheram-Fuller, Locke, & Gulsrud, 2012) to determine the effects of the program on social connectivity. Due to time constraints, Friendship Survey data were only collected for two of the four participants, with data suggesting improvements in social connectivity for participants for whom data were collected. Although the study demonstrated the efficacy of the program in increasing social engagement of participants and parent reports of social functioning, the authors described the results as preliminary due to limitations in design (i.e., failure to demonstrate intervention effects at three different points in time; Kratochwill et al., 2010), a lack of follow-up data, and the lack of Friendship Survey data for half of the participants.

Purpose of the Study

Given the need for effective social skills training interventions for use within school settings, the purpose of the present study was to further evaluate the efficacy of the Superheroes Social Skills program when implemented within a school setting. Given the limitations reported by Radley, Ford, et al. (2014), the current study addresses the reported limitations and serves as a systematic replication and extension of the previous research. The following research questions were generated to guide the investigation:

Research Question 1: Is there evidence of a functional relationship between implementation of the Superheroes Social Skills program and an increase in social engagement of participants with ASD during generalized recess periods?

Research Question 2: Is there a change in sociometric standing of participants with ASD following participation in the Superheroes Social Skills program?

Research Question 3: Do teachers describe participants' social skill use as improving following participation in the Superheroes Social Skills program?

Research Question 4: Do participants with ASD describe the Superheroes Social Skills program as an acceptable intervention?

Method

Participants

Participants were recruited from three public schools in the Southeastern United States and were referred by teachers

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and administrators for participation in the study. Both participants with ASD and typically developing peers were recruited for inclusion in social skills groups. Inclusion criteria for children with ASD consisted of a special education classification of autism or a diagnosis of ASD, educational placement in an inclusive classroom, and no current participation in any type of social skills intervention. However, inclusion criteria allowed for children with ASD who received special education services and had social communication goals not currently being addressed through systematic intervention. Informed consent and assent was obtained from parents and participants prior to collection of baseline data. Inclusion criteria for typically developing peers consisted of teacher nomination due to demonstration of appropriate social skills within the classroom and a neutral or positive history of interaction with the participants with ASD. Informed consent and assent was also obtained from typically developing peers who participated in the groups. Pseudonyms are used throughout the article for participants.

Participants with ASD and typically developing peers were divided between four social skills training groups based on school and grade, each group being comprised of at least one participant with ASD and one typically developing peer. Jacob, a 9-year-old Caucasian male in fourth grade, and Carl, a 10-year-old African American male in fourth grade, were included in Group 1. Jacob had a special education classification of Speech or Language Impairment and a diagnosis of ASD. He was reported to have difficulty engaging in reciprocal conversation, appropriately joining interactions, and requesting assistance from peers. Jacob was also reported to have difficulty understanding others' humor and intentions. Jacob's teacher reported that he frequently made socially inappropriate comments, and occasionally became loud and disruptive following changes in routine. At the time of the study, he received additional pull out special education services for speech (i.e., pragmatic language). Jacob did not receive any special education services for academics. Carl had a special education classification of autism. He was reported to have difficulty with joining in with peers during recess activities, talking about interests of others, using appropriate nonverbal communication skills, and taking turns during games. In addition, Carl's teacher reported that he engaged in repetitive verbal behaviors and patterns of play, and demonstrated difficulty recognizing emotions of other students. At the time of the study, he was receiving pull out services for speech (i.e., articulation, pragmatic language). He did not receive special education services for academics. Group 1 included one typically developing African American male peer, also in the fourth grade.

Edgar, a 9-year-old African American male in third grade, was included in Group 2. Edgar had a special education classification of autism and a diagnosis of ASD. He was reported to have difficulty establishing and maintaining eye contact during conversations, joining in peer groups when invited, acknowledging interests of others, and joining conversations or activities without interrupting. Teacher reports of behavior also indicated that Edgar was frequently manipulated by peers and preferred to engage in solitary activities. Edgar had previously received school-based counseling services to address problem behaviors within the classroom (i.e., inappropriate vocalizations, following directions, staying in assigned seat, completing assignments). At the time of the study, he received additional pull out special education services for speech (i.e., articulation). In addition, Group 2 included two typically developing third grade peers: an African American male and an African American female.

Dylan, an 11-year-old African American male in fourth grade, was included in Group 3. Dylan had a special education classification of autism and a diagnosis of ASD. He was reported to have difficulty engaging with peers during unstructured periods, recognizing nonverbal cues of others, joining interactions without interrupting, and respecting personal space of others. In addition, Dylan engaged in body rocking and repetitive movements, and became disruptive following changes in routine. Dylan's teachers reported that he demonstrated grade-appropriate academic skills. At the time of the study, he received additional pull out special education services for speech articulation and school counseling services for anxiety management. Group 3 also included two typically developing fourth grade peers: a biracial female and a Hispanic male.

Louis, a 5-year-old Caucasian male in kindergarten, was included in Group 4. Louis had a special education classification of developmental delay and a diagnosis of ASD. He was reported to have difficulty initiating social interactions with peers, responding appropriately to invitations to join activities, utilizing eye contact, and requesting information from peers. In addition, Louis's teachers reported use of repetitive and stereotyped phrases and frequent engagement in solitary activities in the presence of peers. He received additional pull out special education services for speech articulation. He did not receive special education services for academics. Two additional peers were included in Group 4. The peers were both Caucasian males and classmates of Louis.

Settings

Participants in Group 1 attended a small elementary school in a small metropolitan area of a Southeastern state. For the school, 80% of the student population was African American, 12% was Hispanic, and 3% was Caucasian. Free or reduced lunch was received by 97% of the student population. The participants from Group 2 and Group 3 attended a small elementary school in a small metropolitan area of a

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Southeastern state, of which 88% of the student population was African American, 9% was Hispanic, and 2% was Caucasian. Ninety-nine percent of the student population was receiving free or reduced lunch. Last, the participants in Group 4 attended a small elementary school in a rural area of a Southeastern state. The student population for the school was 51% Caucasian and 48% African American. Free or reduced lunch was received by 76% of the student population.

Social skills training and observations of social engagement occurred at each of the schools at which participants were enrolled. Social skills training for Group 1 was conducted in a small room used by the school for speech-language services, which contained a round table and chairs. Social skills training sessions for Group 2 and Group 3 were conducted an empty classroom with a small table, chairs, and a dry erase board. Group 4 received social skills training in a small room used for school meetings and storage. The room contained a small round table, chairs, and a dry erase board. Observations of social engagement occurred during recess periods for each participant. Recess facilities were similar across participants, with children having access to a grass field and playground equipment. Group 3, however, did not have access to playground equipment and was provided with jump ropes, balls, and other recess equipment.

Dependent Measures

Social engagement. The primary dependent variable in the current study was social engagement of participants with ASD during 10-min unstructured recess periods. Social engagement data were collected by observers uninvolved with intervention procedures. Although uninvolved in intervention procedures, raters were not blind to study hypotheses. Social engagements of participants were coded using operational definitions from the Playground Observation of Peer Engagement (POPE; Kasari, Rotheram-Fuller, Locke, & Guiou, 2005) in conjunction with a 10-s momentary time sampling (MTS) procedure. Consistent with MTS procedures, observers recorded the state of engagement displayed by the participant with ASD at the conclusion of each 10-s interval. A total of five different states of engagement were coded. Intervals were considered as socially engaged if the participant with ASD was engaged in direct social behavior with peers (e.g., engaging in conversation, cooperative play). Participants with ASD were also coded as socially engaged if they were observed to be participating in an organized activity in which all participants were playing by a prespecified set of rules. Additional engagement states coded included solitary, defined as the child playing alone with no peers within 3 feet; proximity, defined as the child playing alone within 3 feet of peers; onlooker, defined as the child watching peers who are farther away than 3 feet; and parallel, defined as the child and peers engaging in a similar activity but failing to engage. Intervals coded as solitary, proximity, onlooker, and parallel were considered not socially engaged.

Social functioning. Prior to collection of baseline data, social skills to be taught to participants were identified using the Autism Social Skills Profile (ASSP; Bellini & Hopf, 2007). In addition, the ASSP was completed upon the conclusion of the intervention to assess changes in social functioning. General education teachers of each participant completed the ASSP, rating social skill performance of participants, prior to and following conclusion of social skills training. The ASSP provides a Total social functioning score and three subscale scores including Social Reciprocity, Detrimental Social Behaviors, and Participation/Avoidance. The measure utilizes a 4-point Likert-type scale, with a rating of 1 indicating that the participant never engages in the behavior and a rating of 4 indicating that the participant very often engages in the behavior. High scores on the ASSP indicate frequent utilization of socially appropriate behavior. Studies evaluating the psychometrics of the ASSP have found that the measure has high internal consistency ($\alpha = .940$) and test–retest reliability ($\alpha = .904$).

Sociometric status. In addition to the primary dependent variable, sociometric data were gathered to further evaluate the efficacy of the intervention package. The Friendship Survey (Kasari et al., 2012) was administered to determine if there was a change in the participants' social prominence in the classroom following social skills training. During the baseline phase, a trained graduate student administered the Friendship Survey to participants and their classmates in a group format. The Friendship Survey asks students to identify students with whom they like to "hang out," and to identify other groups of children who "hang out" together. The measure provides information about the social network salience (SNS) of an individual student in the class. SNS scores were calculated using procedures developed by Cairns and Cairns (1994). SNS scores allow for determination of the social prominence of participants with ASD in relation to peers. Participants' SNS are interpreted using the following metric: isolated (0), peripheral (1), secondary (2), or nuclear (3). A minimum of 50% of children in each classroom must complete the Friendship Survey to ensure reliability of data (Cairns & Cairns, 1994). Previous evaluations of the Friendship Survey indicate high test–retest reliability ($r = .74$–$.84$) and concurrent validity with observational data (Cairns, Leung, Buchanan, & Cairns, 1995).

Acceptability. Acceptability of the intervention was assessed using a modified version of the Children's Intervention Rating Profile (CIRP; Witt & Elliott, 1985). The CIRP was administered to participants and typically developing peers

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by a trained graduate student following the completion of social skills training. The CIRP utilizes a 6-point Likert-type scale, with a rating of 1 indicating that the participant agrees with the item and a rating of 6 indicating that the participant does not agree with the item. Samples of items on the CIRP include, "I liked Superheroes Social Skills," "Superheroes Social Skills may cause problems with my friends," and "Superheroes Social Skills would be a good method to use with other children." Studies have found the internal consistency of the CIRP to range from .75 to .89 (Elliott, 1986).

Procedures

Design. A concurrent multiple baseline across participants and settings design (Cooper, Heron, & Heward, 2007) was utilized to evaluate the efficacy of the social skills training package on social engagement behaviors of participants with ASD during recess periods. The study consisted of three phases: baseline, intervention, and follow-up. Following demonstration of stable, low levels of engagement for participants in Group 1, social skills training commenced. Following demonstration of increased mean levels of social engagement across two consecutive data points for Group 1 in conjunction with a decreasing trend of social engagement for Edgar, social skills instruction began for Group 2. Similarly, introduction of intervention for Group 3 followed observation of increased social engagement across two data points for Edgar and stable, low levels of social engagement for Dylan. Last, implementation of intervention for Group 4 followed observation of increased social engagement across two data points for Dylan in conjunction with stability in engagement data for Louis.

Baseline. During the baseline phase, the teachers completed the ASSP for each participant with ASD to identify skills to be taught and provide a pre-intervention measure of social functioning. In addition, the Friendship Survey was administered to participants with ASD and their classmates prior to the collection of direct observations. Direct observations of social engagement occurred during participants' scheduled recess periods. Participants had access to playground equipment and/or a field, and no restrictions were placed on the participants during baseline data collection. Recess setting and access to recess equipment was consistent across observations. Trained graduate students conducted direct observations of social engagement from an unobtrusive location on the playground to minimize the possibility of reactivity.

Intervention. Social skills training sessions were conducted in small groups facilitated by trained graduate students. Social skills training sessions were approximately 30 min in length and occurred once weekly. Using ASSP data, a total of four social skills from the Superheroes Social Skills program were identified to be taught across all groups: Participate (i.e., joining in an activity), Body Basics (i.e., nonverbal communication skills), Turn Taking, and Conversation/Topic Maintenance. A new skill was introduced only after target students reached mastery of the previous skill. Mastery of target social skills was assessed using an intervention manual-derived acquisition checklist during role-plays conducted at the end of each social skills training session. Mastery of skills was also confirmed immediately prior to instruction of a new target skill. Due to some participants meeting mastery criterion sooner than others, total duration of intervention ranged from 7 to 11 weeks.

Social skills training sessions followed the manualized Superheroes Social Skills program format. Readers are referred to Radley, Jenson, Clark, Hood, and Nicholas (2014) for a detailed description of implementation of the program. Each intervention session began by reviewing the group rules and schedule for the session. Next, the target social skill was introduced via DVD by animated superheroes. Introduction to the target social skill by animated superheroes was approximately 3 min in duration. Following the introduction of the skill, participants watched video models of unknown peers utilizing the target skill appropriately, followed by in vivo modeling of the skill by the facilitator and typically developing peers. Participants were then provided with a self-monitoring card and engaged in three to four role-plays using the target social skill with peers. During role-plays, correct demonstration of the target skill was followed by facilitator praise and encouragement to utilize the self-monitoring card. Corrective feedback was provided following observations of incorrect use of the skill. Following role-plays and self-monitoring, participants watched an animated social story, which provided a review of steps for demonstration of the skill. Participants then engaged a social game that required use of the target skill. At the conclusion of each group, participants were provided with a reward for participation in the group (e.g., small tangible). In addition, participants were instructed by the facilitator to continue to utilize the self-monitoring card in generalized settings.

During the intervention phase, direct observation of social engagement occurred approximately once a week during participants' recess periods, with observations conducted by trained graduate students using the same procedures as during baseline observations. Recess conditions were the same as in the baseline condition with participants having access to playground equipment and a field. No restrictions were placed on participants during intervention phase recess periods. At the conclusion of the intervention phase, teachers completed the ASSP for a second time, participants with ASD and typically developing peers included in social skills groups completed the CIRP, and the Friendship Survey was administered to participants and their classroom peers.

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Follow-up. Follow-up data were collected 5 weeks after the conclusion of the intervention phase. Procedures during follow-up were identical to those during baseline. A minimum of three follow-up observations of social engagement during recess periods were conducted for each participant with ASD. At the conclusion of the follow-up phase, participants with ASD and their classroom peers again completed the Friendship Survey. As a result of end of the school year time constraints, follow-up data were not able to be collected for Louis.

Interobserver agreement (IOA). Prior to collection of baseline observations of social engagement, observers participated in training sessions with the primary researcher to establish IOA. During training sessions, the primary researcher and observers reviewed state of engagement codes from the POPE. Observation training videos, which depicted unknown children in a play setting, were used to practice utilization of the coding system. Practice observations were repeated until a minimum IOA of 90% was established between the primary researcher and observers.

Throughout the duration of the study, a trained observer coded social engagement of participants with ASD simultaneously with another trained observer to assess IOA. IOA was obtained during a minimum of 33% of data points per participant per phase. IOA was calculated by dividing the number of agreements by the number of agreements and disagreements and multiplying by 100 to obtain a percentage. Mean IOA for Jacob was 91% (range = 82%–97%), 93% for Carl (range = 88%–97%), 89% for Edgar (range = 83%–100%), 90% for Dylan (range = 84%–100%), and 94% for Louis (range = 87%–100%). In addition to simple IOA, kappa was also calculated. Kappa accounts for both occurrences and non-occurrences of behavior (Sattler & Hoge, 2006), providing a better estimate of agreement for states of engagement than simple IOA. Kappa was calculated using the formula provided by Uebersax (1982). Kappa was found to be 0.84 (95% CI = [0.77, 0.91]) for Jacob, 0.87 (95% CI = [0.82, 0.93]) for Carl, 0.80 (95% CI = [0.70, 0.89]) for Edgar, 0.90 (95% CI = [0.85, 0.97]) for Dylan, and 0.89 (95% CI = [0.84, 0.94]) for Louis. All kappa values represent good to very good agreement.

Intervention integrity. To evaluate the extent to which the social skills training sessions were implemented with integrity, researchers completed an intervention-derived treatment integrity checklist after 100% social skills training sessions. The percentage of steps accurately completed by the researchers during each session was calculated. Intervention integrity was found to be 100% across all groups. IOA for intervention integrity was collected during a minimum of 33% of data points by an independent observer and was found to be 100%.

Data analysis. The efficacy of the intervention was evaluated primarily through visual analysis of social engagement data. Six data features were utilized to determine the effect of the intervention on social engagement of participants with ASD: level of data, trend of data, variability of data, immediacy of intervention effect, overlap between phases, and consistency of data patterns across participants (Kratochwill et al., 2010). In addition, effect sizes were determined by calculating NAP (Parker & Vannest, 2009). NAP is a nonparametric statistic used to determine the magnitude of differences between phases by evaluating the overlap between each baseline phase data point and each intervention phase data point. NAP is strongly correlated to R^2, making it particularly useful for evaluation of single case data (Parker & Vannest, 2009). NAP scores range from 0.00 to 1.0, with scores from 0.00 to 0.65 being considered as weak effects, scores from 0.66 to 0.92 being considered moderate effects, and scores from 0.93 to 1.00 being considered strong effects (Parker & Vannest, 2009). Finally, descriptive statistics were used to evaluate changes in SNS and social functioning, as well as to evaluate the acceptability of the intervention.

Results

Social Engagement

The primary dependent variable for the current study was social engagement during the recess period. These data were collected using the procedures previously described. Engagement state data for each participant are presented in Figure 1. An omnibus effect size was calculated for baseline to intervention phase comparisons for all participants (NAP = 0.95; 95% CI = [0.65, 1.25]), indicating a strong intervention effect. Follow-up data collected for four participants indicated that the intervention effects were maintained 4 to 6 weeks after the intervention was terminated, with an omnibus effect size indicating a strong maintenance effect (NAP = 0.97; 95% CI = [0.51, 1.42]).

Jacob was socially engaged during an average of 8% of intervals during baseline. Once the intervention was implemented, immediate increases in level and trend were seen, with social engagement occurring during 37% of intervals in the intervention phase (NAP = 0.93, 90% CI = [0.19, 1.51]). Jacob was socially engaged during an average of 34% of intervals during follow-up observations (NAP = 1.00, 90% CI = [0.16, 1.84]), again demonstrating a strong intervention effect. Carl demonstrated a low, stable level of social engagement data during baseline, and was socially engaged during an average of 6% of intervals. Although his intervention data were variable, a noticeable change in level is present as Carl was socially engaged during an average of 38% of intervals during the intervention phase (NAP = 1.00, 90% CI = [0.36, 1.64]). Carl demonstrated increased

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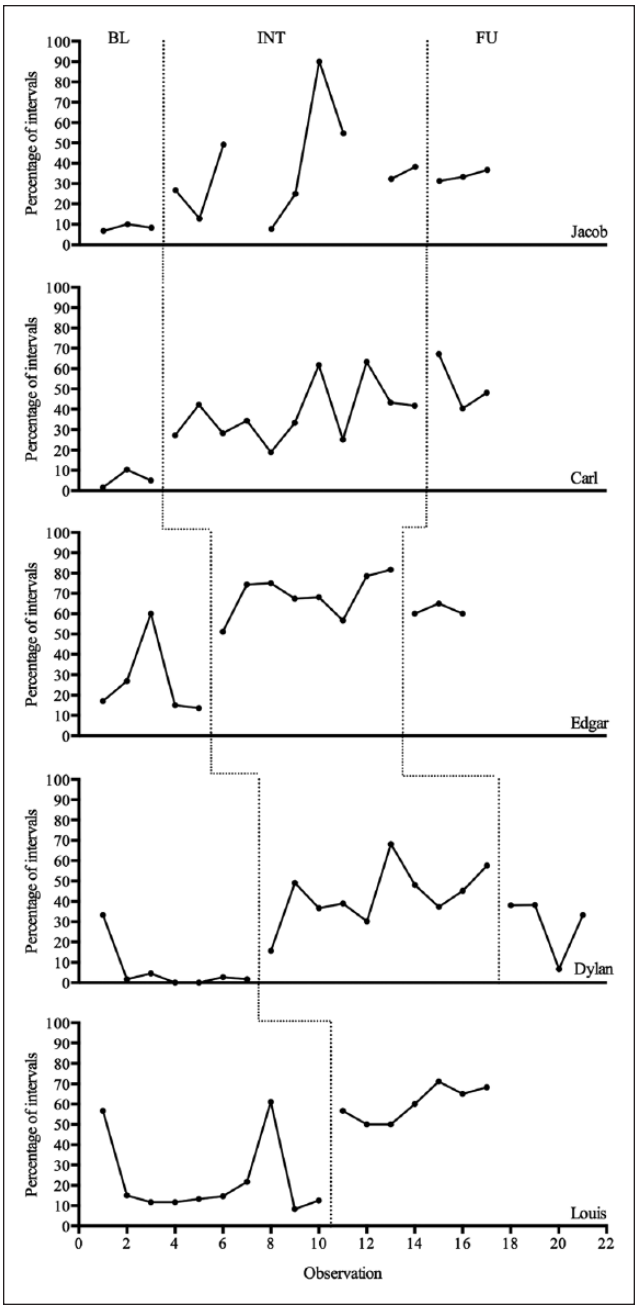


Figure 1. Percentage of intervals of social engagement during recess periods.
Note. Missing data points for Jacob indicate absences from school during recess observations. BL = baseline, TNT = intervention, FU = follow-up.

social engagement during the follow-up observations, with social engagement observed during 52% of intervals (NAP = 1.00, 90% CI = [0.16, 1.83]).

Edgar's baseline data were low but variable, with social engagement being observed during an average of 27% of intervals. Upon implementation of the intervention, an immediate, stable increase was observed, as Edgar was socially engaged during an average of 69% of intervals (NAP = 0.95, 90% CI = [0.34, 1.46]). This increase was maintained in the follow-up phase, where Edgar was observed to be socially engaged during 62% of intervals (NAP = 0.93, 90% CI = [0.13, 1.60]). Dylan demonstrated relatively stable and low baseline data (with the exception of Datum Point 1), with an average of 6% intervals of social engagement. Dylan demonstrated an immediate increase in level with an upward trend during intervention, and was observed to be socially engaged during an average of 43% of intervals during the intervention phase (NAP = 0.97, 90% CI = [0.46, 1.42]). During the follow-up observations, Dylan was observed to be socially engaged during 29% of intervals (NAP = 0.95, 90% CI = [0.27, 1.52]). Louis demonstrated a low but somewhat variable level of social engagement during baseline, and was observed to be socially engaged during an average of 23% of intervals. Upon implementation of the intervention, there was an immediate increase in level with stable data collected throughout the intervention phase. Louis was observed to be socially engaged during an average of 60% of intervals in intervention (NAP = 0.91, 90% CI = [0.33, 1.30]). Due to the end of the school year, no follow-up data were collected for Louis.

Social Functioning

The pre- and post-intervention ASSP results for each participant are presented in Table 1. With the exception of Jacob, the teachers' responses on the ASSP completed after the intervention was terminated indicated improved overall social skills when compared with responses on an ASSP completed prior to implementation of the intervention. Two of five participants were found to improve on the Social Reciprocity subscale, whereas four of five participants improved on both the Social Participation/Avoidance and Detrimental Social Behaviors subscales.

Sociometric Status

SNS was calculated via the Friendship Survey for all participants, as well as their peers. Friendship Surveys were completed prior to intervention, following intervention, as well as after the follow-up phase for all participants except Louis.

Pre-intervention, both Jacob and Carl received a score of 1 on the SNS scale, indicating peripheral social status. Prior to intervention, both Jacob and Carl's SNS scores were lower than the mean SNS score of their classroom peers (SNS = 2.42 and 2.16, respectively). Post-intervention, Jacob and Carl both received friendship nominations that resulted in a SNS of 2, indicating that they made modest progress in terms of peer nomination for group inclusion that was maintained following termination of social skills

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Table 1. ASSP Teacher Report Pre- and Post-Intervention Scores.

	Social reciprocity		Social participation/avoidance		Detrimental social behaviors		Total	
Participant	Pre	Post	Pre	Post	Pre	Post	Pre	Post
Jacob	70	53	41	29	33	26	158	121
Carl	37	46	29	33	31	32	105	122
Edgar	41	48	27	35	24	31	101	123
Dylan	29	25	25	43	32	53	98	130
Louis	30	22	32	35	47	56	119	126

Note. ASSP = Autism Social Skills Profile.

training. Both Jacob's and Carl's SNS score was found to exceed the mean SNS score of their classroom peers (SNS = 1.68 and 1.77, respectively). Completion of the friendship survey during follow-up indicated that Jacob and Carl maintained an SNS score of 2, continuing to exceed the mean SNS of their classroom peers (SNS = 1.89 and 1.66, respectively).

Prior to intervention, both Edgar and Dylan were identified as isolated in their classrooms (SNS = 0), scores that were less than that of their classroom peers (0.88 and 1.18, respectively). Post-intervention, Edgar was found to have made considerable improvements in SNS, moving from the isolated range to being secondary (SNS = 2). Edgar's post-intervention score demonstrated greater improvement than the mean SNS of his classroom peers (SNS = 1.66). Dylan's SNS score was in the peripheral range (SNS = 1) post-intervention, less than the mean SNS of his classroom peers (SNS = 1.33). Whereas Edgar maintained an SNS score of 2 during follow-up, the mean of his classroom peers was found to decrease (SNS = 1.16). Dylan maintained an SNS of 1 during maintenance, which was less than the mean SNS score of his classroom peers (SNS = 1.53).

Pre-intervention, Louis received a score of 0 on the SNS scale. Mean SNS score for classroom peers was also found to be 0. Post-intervention SNS scores indicated no change, with both Louis and peers receiving scores of 0 on the SNS scale. Due to the end of the school year, no SNS data were collected at follow-up for Louis and his classroom peers.

Acceptability

A modified version of the CIRP (Witt & Elliott, 1985) was used to assess participant acceptability of the intervention used in the present study. The rating scale was given to each of the participants with ASD and the typically developing peers. Each of the participants found the intervention to be highly acceptable. The average CIRP score for the five participants with ASD was found to be 4.96. Jacob's mean item rating was 5.33, Carl's mean item rating was 4.33, Edgar's mean item rating was 4.83, Dylan's mean item rating was 4.33, and Louis's mean item rating was 6. The average CIRP score for the typical peers was 5.63 (range = 5–6).

Discussion

The current study evaluated whether a functional relationship existed between participation in the Superheroes Social Skills program within a school setting and increases in social engagement of participants with ASD during recess. Kratochwill et al. (2010) suggested that visual analysis of single case data and subsequent evaluation of functional relationships evaluate level, trend, variability, immediacy of effect, overlap, and consistency of data patterns across similar phases. Following implementation of the intervention, all participants in the study demonstrated immediate increases in level and trend of social engagement during unstructured recess periods. As demonstrated by NAP calculations, data for all participants demonstrated minimal overlap. In addition, consistency in social engagement data during recess periods was observed across participants and phases. Although similar improvements were observed across participants, various degrees of variability in social engagement were observed across participants. Variability is frequently observed in research targeting social engagement of children with ASD (e.g., Bellini, Akullian, & Hopf, 2007; Koegel, Vernon, & Koegel, 2009; Mason et al., 2014), and may be a result of factors such as availability of preferred peers or activities. Although recess facilities were similar across participants, variations in size of recess area may have contributed to variability by allowing for wider dispersal of peers with whom participants with ASD were able to interact. In the case of Jacob, inconsistent attendance may also be associated with increased variability in social engagement.

Overall, data collected in the current study meet Kratochwill et al.'s criteria for demonstration of a functional relationship and indicate that participation in the Superheroes Social Skills intervention resulted in improvements in social engagement behaviors of participants during recess periods—supporting preliminary findings of other studies (e.g., Block et al., 2015; Radley, Ford, et al., 2014). Given that research in social skills training for children with ASD has often been limited by findings of poor generalization of target skills to nontraining settings (e.g., Bellini, Peters, et al., 2007; Rao, Beidel, & Murray, 2008; Reichow & Volkmar, 2010), the results of the current study indicate that the combination

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of multiple intervention strategies in Superheroes Social Skills may provide sufficient behavioral exemplars and reinforcement contingencies (Stokes & Osnes, 1989) to overcome threats to generalization.

The current study also addressed limitations of Radley, Ford, et al. (2014) through collection of data at a 5-week follow-up. For participants available for observation during the follow-up period, social engagement during recess periods was substantially improved over baseline levels. Maintenance data were not collected for Louis due to time constraints associated with the end of the school year. Bellini, Peters, et al. (2007) suggested that maintenance effects for school-based social skills training are typically poor. The findings of maintained improvements in percentage of intervals of social engagement support previous findings that participation in the Superheroes Social Skills program results in sustained improvements in accurate demonstration of target skills (Radley, O'Handley, et al., 2014). Taken together, findings of the current study and Radley, Ford, et al. (2014) indicated that the incorporation of multiple evidence-based social skills training strategies (e.g., Lee et al., 2007; Wang & Spillane, 2009; Zhang & Wheeler, 2011) may effectively address poor maintenance effects observed in meta-analysis of school-based social skills training (e.g., Bellini, Peters, et al., 2007).

The observed intervention effects are similar to those observed in Radley, Ford, et al. (2014), which indicated increased social engagement during recess periods following participation in the social skills intervention, as well as improvements in SNS. Although improvements in SNS were noted by Radley, Ford, et al. (2014), SNS data were only collected for two of four participants. The current study addressed this limitation, collecting SNS data for all participants. Results support those of Radley, Ford, et al., as four of five participants demonstrated improvements in SNS greater than their non-participant classroom peers and maintained improvements through follow-up. For Louis and his peers, scores of 0 for both pre- and post-intervention could signify a lack of understanding the Friendship Survey. Students in kindergarten may lack the skills necessary to read instructions and write the names of their friends or other groups of friends. Therefore, these scores may not be indicative of a lack of improvement but rather that these data are not applicable or meaningful for Louis. Improvements observed in SNS are particularly important, as previous research indicates that social skills training may result in improvements in target skills without appreciable impact on sociometric standing (e.g., Fujiki, Brinton, McCleave, Anderson, & Chamberlain, 2013). These increases suggest that participation in the intervention may have contributed to the formation of social relationships, an important component for long-term outcomes such as academic achievement and employability (Shtayermman, 2007). It is also possible that inclusion of peers in social skills groups contributed to improved SNS of participants with ASD via acquisition of untrained social skills via peer modeling, through establishment of new social relationships between participants with ASD and peers included in the group, and by improving non-participating peer perception of participants with ASD by being increasingly associated with peers who demonstrated appropriate social skills.

The current study found participation in the social skills intervention to result in improved overall social functioning for four of five participants, as rated by teachers. This finding suggests that participants may have generalized newly learned social skills to classroom settings. It is, however, important to note that Jacob's teacher did not indicate improvements in social functioning within classroom despite increased social engagement during recess periods and improved SNS. Several factors may have contributed to a lack of observed improvement on teacher ratings of social skills, such as a high rating at baseline and sporadic absences during intervention. Additional research is required to determine variables that may contribute to a lack of improvement as rated by teachers.

The results of the current study have important implications for school-based social skills training. Although several studies have identified social skills training strategies that may be effective in school settings (e.g., Plavnick, Sam, Hume, & Odom, 2013), challenges exist that limit adoption in such settings. School-based intervention is often limited to manualized interventions (Dingfelder & Mandell, 2011). However, few manualized, practice-ready social skill curricula are available to school personnel (Lord et al., 2005). As a result, evidence-based practices are often neglected for nonempirically supported practices in these environments (Hess et al., 2008). Given the documented lag between the development of evidence-based practices and the adoption of such practices (Walker, 2004), the Superheroes Social Skills program's manualized nature may address a clear need for practice-ready social skills curricula for use within school settings. As the curriculum was facilitated by trained graduate students in the current study, data regarding the acceptability of intervention procedures to teachers were not collected as part of the current study. However, high ratings of child acceptability further indicate the utility of the program as a school-based intervention.

The results of the study indicate that the intervention utilized in the current study may be beneficial for children with ASD, regardless of special education classification by which school-based services are provided. Although a small sample of children with ASD were included in the study, generalization of findings is enhanced through inclusion of participants of diverse ages and social deficits. Generalization of findings is, however, limited to high functioning children with ASD with educational placements in inclusion classes. As an additional limit to generalizability of findings, it must also be noted that an exclusionary criteria of the current study was participation in intervention targeting social functioning at the time of the study. As such, participants with

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ASD included in the current study are not reflective of findings suggesting that approximately 60% of school-age children with ASD currently receive social skills training (Pringle, Colpe, Blumberg, Avila, & Kogan, 2012). However, participants with ASD included in the study were similar to other school-age children with ASD in that speech or language therapy was the primary mode of intervention service and 40% of participants had received intervention targeting problem behaviors (Pringle et al., 2012[).

Although the current study has implications for practitioners in school settings, limitations must be considered. First, the Superheroes Social Skills program incorporates multiple strategies to promote generalization (i.e., peer modeling, role-plays, incorporating typically developing peers, and self-monitoring). The simultaneous use of these strategies limits the degree to which individual components can be identified as functionally related to increases in social engagement. Future investigations should conduct a component analysis of the program, as some of these individual components may be more vital than other components. Furthermore, procedural integrity was not assessed for the self-monitoring component. The group facilitator encouraged participants to self-monitor skill use in generalized settings, but the extent to which this occurred was not measured. Future research may address this limitation through incorporation of peer monitoring of self-monitoring in the classroom setting. A lack of observers blind to study hypotheses should also be considered a limitation of the current study. However, high levels of IOA suggest that observational data may be considered accurate representations of social engagements of participants. Future research should address this limitation through use of blinded observers. Differences in group composition (i.e., Group 1 being comprised of two participants with ASD and one typically developing peer) across groups may be considered a limitation. Although improvements were observed across all participants, future research should investigate the effect of differences in group composition on participant outcomes. No social engagement data were collected for comparison peers. As such, it is unknown how similar or dissimilar levels of social engagement of participants were to other peers present during recess periods. Although future research should address this limitation, prior research indicates that ASD interact with others less frequently than peers (Anderson, Moore, Godfrey, & Fletcher-Flinn, 2004) and improvements in social engagement observed in the current study may be considered clinically significant (e.g., Cappadocia & Weiss, 2011).

Although the typically developing participants in this study may have improved their exhibited use of social skills, or may have sought to include the target students in activities outside of group participation, no direct or indirect measures of typically developing peer models were used. Future research may investigate the effect of participation in the curriculum as a typically developing peer. As previously noted, kindergarten-age participants may have lacked the necessary skills to successfully complete the Friendship Survey. Future studies may consider simplified measures to assess sociometric status with students in lower elementary grades, particularly strategies that do not require the students to read or write. Last, trained graduate students conducted the social skills lessons in pull out sessions. As such, it is unclear whether similar results would be observed if school personnel implemented the intervention. Future research should examine child behavior following participation in the Superheroes Social Skill program when facilitated by school personnel. As research has yet to evaluate implementation of the intervention by school personnel, researchers and practitioners may consider using a model of training similar to that utilized in the current study, namely, facilitation of social skills training in a pull out setting and inclusion of peers who demonstrate appropriate social skills. Although peers included in the current study did not have special education classifications, implementation by school personnel may favor inclusion of peers with disabilities other than ASD (Sansosti, 2010), allowing practitioners to target shared deficits across disability categories. Although the Superheroes Social Skills program is manualized, future research is also required to determine the degree of training required for school personnel of various backgrounds (e.g., special education teachers, school psychologists, classroom or student aides) who may engage in social skills training and support.

Conclusion

Approximately 80% of children with ASD receive school-based social skills intervention (Ruble et al., 2010). Despite the large number of children who receive school-based social skills training, intervention in this context is largely ineffective and often utilizes ineffective intervention strategies (e.g., Bellini, Peters, et al., 2007; Hess et al., 2008; January, Casey, & Paulson, 2011). To improve the quality of services delivered in school settings, school personnel are in need of manualized curricula that address social deficits of children with ASD (e.g., Dingfelder & Mandell, 2011; Lord et al., 2005). Results of the current study provide further support that the Superheroes Social Skills program may be utilized in a school setting, resulting in increases in social engagement of children with ASD that are generalized to nontraining settings. In addition to generalization, effects of participation in the program were observed at a 5-week follow-up. Although future research should address the limitations of the current study, results indicate that implementation of the Superheroes Social Skills program may promote social skill use in high functioning children with ASD in inclusion classrooms.

Declaration of Conflicting Interests

The author(s) declared the following potential conflicts of interest with respect to the research, authorship, and/or publication of this

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article: Keith C. Radley is a coauthor of the Superheroes Social Skills program. The Superheroes Social Skills program was developed for the Utah State Office of Education as part of his doctoral training and is distributed free of cost to Utah educators. He does not benefit financially from the commercial sale of the program, nor were the data included in the article part of research funded by the commercial publisher.

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